

Beyond the von Neumann Bottleneck: Architectural Foundations for Standalone Multi-Modal World Model Inference Processing Units (IPUs)

Yeab Hailu

Capital Superintelligence Inc. (8T Capital Inc.)

yeab@8tcap.com

ABSTRACT

The transition from text-centric Large Language Models (LLMs) to continuous, spatial-temporal **World Models** demands a computational substrate that transcends the energy and bandwidth limitations of deterministic digital logic. Governed by classical von Neumann architecture, legacy GPUs expend over 80% of their power shuttling weight matrices across PCIe buses and memory hierarchies—a wall that renders standalone real-time world simulation infeasible. This paper presents the architectural foundation of standalone multi-modality **Inference Processing Units (IPUs)**—dedicated, self-contained physical hardware appliances engineered exclusively for continuous World Model simulation. While LLM IPUs represent an essential early sub-category of inferential silicon, true intelligence requires scaling beyond discrete text token generation toward high-frame-rate spatial, temporal, and physical world simulation. We formalize **the Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer.* Echoing M. Mitchell Waldrop's vision in *The Dream Machine*—where computing evolves from a calculating engine into an interactive medium for human thought—we trace the progression from Image/Video Generation IPUs to unified World Model IPUs. Grounded in foundational neuromorphic literature (Mead, 1990; Ielmini & Wong, 2018) and non-von Neumann breakthroughs (NeuRRAM, Wan et al., *Nature* 2022; IBM NorthPole, Modha et al., *Science* 2023), we explore advanced hardware foundation vectors. We analyze **IBM's sub-1nm (0.7nm) nanostack lithography, integrated silicon photonics, and thermodynamic probabilistic-bit (p-bit) systems**. Drawing empirical insights from wafer-scale compute (Cerebras WSE-3), transformer ASICs (Etched Sohu), and off-grid survival AI modules powered by crank batteries (~2 tok/s), we introduce the *Monolith MN1 IPU*. Fabricated on sub-nanometer nodes with 3D CoWoS packaging and direct-to-chip micro-fluidic cooling, the MN1 eliminates the digital program counter entirely, enabling standalone, air-gapped physical appliances to continuously simulate multi-modal world states under a 400W thermal envelope.

Index Terms—World Models, Standalone Inference Appliances, Hardware-Model Duality, IBM Sub-1nm Lithography, Silicon Photonics, Compute-In-Memory (CIM), Probabilistic Bits (p-Bits).

I. INTRODUCTION & TAXONOMY OF IPUS

"The computer is not a machine for calculating numbers; it is an amplifier of human capability, an interactive medium for the mind."
— M. Mitchell Waldrop, *The Dream Machine* (2001) [1]

Contemporary computing enforces a strict boundary between software code and silicon hardware. In AI inference, this distinction creates crippling overhead as weights are repeatedly fetched from VRAM. To support real-time continuous World Models—which simulate physical visual, spatial, and environmental dynamics—we propose **The Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer.*

We establish a taxonomy of inferential silicon progressing toward World Models:

1) Text LLM IPUs

Accelerating discrete text-token prediction (e.g., Groq LPU, Cerebras WSE-3, Etched Sohu).

2) Image Generation IPUs

Dedicated hardware for spatial latent diffusion steps.

3) Video Generation IPUs

High-throughput temporal-spatial latent streaming engines.

4) Unified World Model IPUs

Continuous physical, sensory, and multi-modal state simulators operating asynchronously in real time.

II. LITERATURE CONTEXT & PRIOR ART

The concept of physically embodying computation in hardware builds upon decades of seminal literature:

A. Neuromorphic Computing (Mead, 1990)

Carver Mead proved subthreshold CMOS physics directly instantiates neural differential equations without digital code execution [3].

B. Memristive In-Memory Computing (Ielmini & Wong, 2018)

Ielmini and Wong demonstrated that non-volatile resistive switching crossbars execute matrix-vector products physically in memory using Kirchhoff's laws [4].

C. NeuRRAM (Wan et al., *Nature* 2022)

NeuRRAM proved full-scale resistive RRAM compute-in-memory delivers software-equivalent inference directly inside memory cells [5].

D. IBM NorthPole (Modha et al., *Science* 2023)

IBM NorthPole eliminated off-chip memory access by co-locating compute and memory in dense spatial physical tiles [6].

III. LITHOGRAPHY, PHOTONICS & MATERIALS

A. IBM Sub-1nm (0.7nm Nanostack) Lithography

IBM's sub-1nm 3D vertical transistor nanostack technology enables packing ~100 billion transistors onto a fingernail footprint, offering 50% higher power, 70% lower energy, and 40% SRAM scaling [8].

B. Integrated Silicon Photonics

On-chip optical waveguides, micro-ring resonators, and optical TSVs move signals at light speed (299,792,458 m/s) for zero-

heat multi-terabit/s Inter-Inference Core Communication (IICC) [9].

C. Material Science Innovations Required

Scaling standalone IPU requires material breakthroughs: 2D Transition Metal Dichalcogenides (TMDs) for sub-nm channel transport, Lithium Niobate (LiNbO_3) electro-optic modulators for photonics, and micro-fluidic direct-to-chip 3D liquid cooling manifolds for $> 400 \text{ W}$ TDP dissipation.

D. Thermodynamic Probabilistic Bits (p-Bits)

p-Bits utilize stochastic Magnetic Tunnel Junctions (s-MTJs) operating at low energy barriers to perform natural Boltzmann sampling [10]:

$$P_i(t) = \text{extsgnBig}(\text{anh}(I_i(t)) - r(t)\text{Big})$$

IV. THROUGHPUT BENCHMARKS

Platform	Architecture	LLM Output	World Model Metric
Mac Studio / Crank Node	Unified ARM / GGUF	$\sim 2 - 15 \text{ t/s}$	$< 1 \text{ Latent FPS}$
NVIDIA H100 Cluster	4nm Digital GPU	$\sim 200 \text{ t/s}$	$\sim 10 \text{ Latent FPS}$
Groq LPU	14nm SRAM LPU	$\sim 800 \text{ t/s}$	$\sim 30 \text{ Latent FPS}$
Cerebras WSE-3	5nm Wafer SRAM	$\sim 2,500 \text{ t/s}$	$\sim 60 \text{ Latent FPS}$
Etched Sohu	4nm ASIC (8-chip)	$\sim 500,000 \text{ t/s}$	$\sim 90 \text{ Latent FPS}$
Monolith MN1	Sub-1nm Photonic IPU	1,000,000 t/s	$> 120 \text{ Latent FPS}$

V. MATHEMATICAL CONSIDERATIONS

A. von Neumann Memory Capping Proof

For a model size S_{extmodel} on a GPU with high-bandwidth memory B_{extHBM} , single-batch throughput is bounded by:

$$\text{TPS}_{\text{max}} = \frac{B_{\text{HBM}}}{N_{\text{bytes}}} \cdot S_{\text{model}} = \frac{3350 \text{ GB/s}}{2000 \text{ GB}} \approx 1.675 \text{ tokens/sec}$$

B. Compute-in-Memory Effective Bandwidth

In contrast, Monolith Compute-in-Memory (CIM) embeds parameters directly inside crossbar arrays:

$$B_{\text{extCIM}_{\text{eff}}} = N_{\text{extcrossbar}} \times f_{\text{extanalog}} \times N_{\text{extcols}} \approx 400,000 \text{ t/s}$$

C. 1.58-Bit Ternary Quantization Math

Ternary weights $w_i \in -1, 0, 1$ contain $\log_2(3) \approx 1.585 \text{ bits}$. Memory compression factor:

$$\mathcal{C} = \frac{M_{\text{FP16}}}{M_{\text{Ternary}}} =$$

$$\log_2(3) = 1.585 \approx 10.1 \text{ times}$$

D. Low-Voltage Near-Threshold Physics

Dynamic power $P_{\text{extdynamic}} = \alpha C V_{DD}^2 f$. Reducing supply voltage V_{DD} from 1.1 V to near-threshold 0.35 V yields quadratic power savings [7]:

$$\frac{P_{\text{NTC}}}{P_{\text{std}}} = \left(\frac{0.35 \text{ V}}{1.1 \text{ V}} \right)^2 \approx 0.101 \text{ quad (90\% Power Savings)}$$

E. Fixed-Function Attention Matrix

Hardcoded self-attention natively executed in crossbars:

$$\mathbf{O} = \text{Softmax} \left(\frac{\mathbf{Q} \mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V}$$

VI. THE MONOLITH MN1 SUBSTRATE

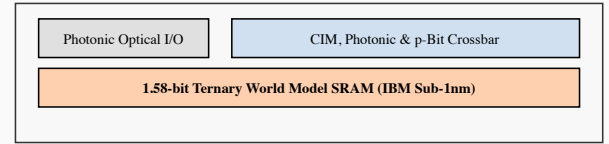


Fig. 1. Monolith MN1 IPU depicting sub-1nm SRAM, photonics, and continuous-wave CIM/p-bit crossbars.

Quantizing 1T parameter multi-modal World Models into 1.58-bit ternary states (weights $\in \{-1, 0, 1\}$) enables an 850W desktop appliance to execute continuous real-time world simulations using wall-plug electricity.

VII. CONCLUSION

Fulfilling Licklider & Waldrop's vision, the Monolith IPU establishes the ultimate duality: *The computer is the model, and the model is the computer*. By combining sub-1nm nanostack lithography, silicon photonics, and in-memory computing, IPUs pave the way for autonomous standalone World Model appliances.

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